



Monetary Authority of Singapore

Quantum Security Questionnaire

April 2026

Quantum Security Questionnaire



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1. Executive Summary

In February 2024, MAS issued an advisory on addressing cybersecurity risks associated with quantum computing¹. The advisory emphasised the need for financial institutions (FIs) to prepare for the advent of cryptographically relevant quantum computers (CRQCs) that could compromise existing encryption and digital signature algorithms. Early preparation is essential not only for the eventual arrival of “Q-day”² but also to ensure resilience should a vulnerability or weakness be discovered in existing traditional cryptographic algorithms.

This questionnaire has been developed to assess the current quantum security preparedness of FIs and understand their future transition plans. The information collected will be used solely to gauge industry readiness and identify collective needs. If required, we may present aggregated and anonymised data at industry sharing. We have also referenced CSA’s Quantum Readiness Index in this questionnaire (<https://form.gov.sg/68d9f7d3a319fad5167c790f>).

This questionnaire is focused on FIs’ quantum security initiatives in Singapore. Please submit your completed questionnaire by **8 May 2026**.

For enquiries regarding this questionnaire, please contact aman_singhal@mas.gov.sg and naomi_sago@mas.gov.sg, with email subject: “[<FI name>] Quantum Security Questionnaire”.

We estimate that it will take approximately 45 minutes to complete this form, excluding time to gather any necessary information. You may save this form and resume later.

¹ <https://www.mas.gov.sg/regulation/circulars/advisory-on-addressing-the-cybersecurity-risks-associated-with-quantum>

² “Q-day” refers to the point at which quantum computers become capable of breaking widely used public-key cryptographic algorithms, rendering current encryption and digital signatures vulnerable. It is not a specific predicted date but a risk planning construct to guide timely migration to quantum-safe cryptography.



2. FI Profile and Contact Details

2.1 FI Details

- i. FI Name: _____
- ii. Please indicate the **primary** financial sub-sector your FI is in.
 - a) Banking
 - b) Capital Markets
 - c) Financial advisory
 - d) Insurance
 - e) Payments
 - f) Others: _____

2.2. Contact Information for Quantum Security Matters

Primary Contact

- Name: _____
- Title: _____
- Department: _____
- Email: _____

Alternative Contact

- Name: _____
- Title: _____
- Department: _____
- Email: _____



3. Quantum Security Questionnaire

Keeping abreast of the latest developments in quantum computing, and raising awareness of the associated cybersecurity risks

3.1 Monitoring and Awareness of Quantum Developments

Question 1. How frequently does your FI monitor developments in quantum computing and associated cybersecurity risks?

- i. At least once a month
- ii. Quarterly
- iii. Semi-Annually
- iv. Annually
- v. Ad-hoc basis

Question 2. Is your board appraised of quantum-related cybersecurity risks?

- i. Yes
- ii. Not yet

Question 3. Has your senior management been briefed on quantum computing cybersecurity risks?

- i. Yes, comprehensive briefing conducted
- ii. Yes, basic awareness session held
- iii. No, but planned within next 12 months
- iv. No current plans

Question 4. Which quantum safe solutions have you evaluated? (Select all that apply)

- i. Post Quantum Cryptography (PQC)
- ii. Quantum Key Distribution (QKD)
- iii. Hybrid quantum cryptography (quantum + classical)
- iv. None evaluated yet



v. Others: _____

Question 5. Which quantum-safe solutions does your FI plan to prioritise in your migration strategy?

(Select all that apply)

- i. Post Quantum Cryptography (PQC)
- ii. Quantum Key Distribution (QKD)
- iii. Hybrid quantum cryptography (quantum + classical)
- iv. Others: _____

Question 6. Which sources does your FI currently use to stay informed about quantum computing developments? (Select all that apply)

- i. Academic research
- ii. Industry reports
- iii. National cybersecurity agencies
- iv. Financial sector regulators
- v. Vendor communications
- vi. International standards bodies, e.g. NIST, ETSI
- vii. Industry associations, e.g. ISACs
- viii. Cybersecurity conferences and events
- ix. Others: _____

Question 7. What industry groups, research bodies, or ISACs do you participate in or engage with for quantum security collaboration? (Select all that apply)

- i. Industry consortiums, e.g. FS-ISAC, ABS-SCCS, iSCCS
- ii. Government-led quantum security initiatives
- iii. University Researchers
- iv. Subject Matter Experts/ Consultants
- v. No formal participation in quantum-focused groups
- vi. Others: _____



3.2 Governance and Leadership

Question 8. How does your FI view quantum risks vis-à-vis existing cybersecurity risks?

- i. The FI does not view the quantum risks as a risk to be addressed in the near term.
- ii. The FI acknowledges that the quantum risk needs to be addressed and is starting to plan its quantum strategy.
- iii. The FI has developed their strategy and started monitoring quantum threats on a regular basis.
- iv. The FI continuously evaluates and tracks the effectiveness in managing quantum risk.

Question 9. Do you have a designated quantum security champion or steering committee?

- i. Yes
- ii. No

3.3 Supply Chain Risk Assessment and Vendor Management

Question 10. Have you assessed quantum-related risks in your IT supply chain?

- i. Comprehensive assessment completed
- ii. Partial assessment conducted
- iii. Assessment planned within next 12 months
- iv. No assessment plans at the moment

Question 11. Have you engaged with your critical IT vendors regarding quantum-resistant solutions?

- i. Yes, formal discussions initiated with all critical vendors
- ii. Yes, discussions with some key vendors
- iii. Preliminary inquiries made
- iv. No engagement yet

Question 12. Do your new or upcoming vendor contracts include provisions for quantum-resistant solution migration?

- i. Yes, all new contracts include such provisions
- ii. Yes, some contracts include provisions
- iii. Under consideration for future contracts



- iv. No provisions currently included

Question 13. Do you track quantum readiness of your vendors and if so, have you identified alternative vendors or solutions for critical systems that may not be quantum-ready in time?

- i. Yes tracking, alternative vendors identified for all critical systems
- ii. Yes tracking, alternatives identified for some critical systems
- iii. Yes tracking, but have not identified alternative vendors or doing so in ad-hoc manner
- iv. Not tracking but currently evaluating alternative options in an ad-hoc manner
- v. Not tracking nor evaluating alternatives at the moment

Question 14. What criteria do you use when evaluating new vendors regarding quantum security capabilities for critical systems?

- i. Quantum-resistant roadmap is a mandatory requirement
- ii. Quantum preparedness is a key evaluation factor
- iii. Quantum security is considered but not weighted heavily
- iv. Currently developing quantum-related vendor criteria
- v. Quantum security not yet part of vendor evaluation



Maintaining an inventory of cryptographic assets, and identifying critical assets to be prioritised for migration to quantum-resistant encryption and key distribution.

3.4 Cryptographic Asset Discovery and Documentation

Question 15. Have you completed an inventory of cryptographic assets used across your FI?

- i. Comprehensive inventory completed and maintained
- ii. Partial inventory completed
- iii. Inventory in progress
- iv. Not yet started

Question 16. If you have not completed the inventory, what is your timeline to do so?

- i. In less than 6 months
- ii. In 6 to 12 months
- iii. 1-2 years
- iv. 2-5 years
- v. Beyond 5 years
- vi. No plans yet
- vii. Others: ____

Question 17. For your cryptographic inventory, do you maintain information on: (Select all that apply)

- i. Cryptographic algorithms and key lengths used
- ii. Ownership responsibilities
- iii. Specific systems or applications where algorithms are embedded
- iv. Expiry date for certificates and key rotation deadlines
- v. End-of-Life dates for cryptographic algorithms (encryption algorithms, hashing algorithms etc)
- vi. Vendor dependencies for each cryptographic solution
- vii. Quantum vulnerability assessment (high/medium/low risk)
- viii. Integration dependencies with other systems



Question 18. How frequently do you update your cryptographic asset inventory?

- i. Continuously/real-time updates
- ii. Monthly
- iii. Quarterly
- iv. Annually
- v. Ad-hoc basis only

Question 19. Have you determined if your cryptographic assets are vulnerable to quantum risks?

- i. Yes
- ii. Not yet determined

Question 20. What criteria do you use for prioritising assets for quantum-resistant migration? (Select all that apply)

- i. Data classification levels
- ii. System criticality
- iii. Risk exposure assessment
- iv. Data retention periods
- v. Regulatory requirements
- vi. Customer impact potential
- vii. Costs involved
- viii. Vendor roadmap availability
- ix. Interdependencies with other systems
- x. Others: _____

Question 21. If no classification has been undertaken, please share the factors that are preventing or delaying the implementation of quantum risk asset classification (Select all that apply)

- i. Strategic uncertainty - Unclear threat timeline or business impact assessment incomplete
- ii. Resource constraints - Insufficient budget, personnel, or technical expertise allocated
- iii. Technology readiness - Waiting for standardisation or maturity of quantum-resistant solutions
- iv. Organisational readiness - Governance, risk management, or change management processes not yet established



- v. External dependencies - Awaiting vendor solutions, regulatory guidance, or industry coordination
- vi. Competing priorities - Other cybersecurity or digital transformation initiatives taking precedence
- vii. Risk tolerance - Current threat level assessed as acceptable for business operations

3.5 IT Infrastructure Crypto-Agility Assessment

Question 22. Crypto-agility refers to the ability to quickly and efficiently migrate from vulnerable cryptographic algorithms to quantum-resistant alternatives. How have you assessed your IT infrastructure's readiness for this transition?

- i. Comprehensive assessment completed across all systems
- ii. Assessment completed for critical systems only
- iii. Assessment in progress
- iv. Not yet assessed

Question 23. What IT infrastructure limitations have you identified that may hinder quantum security transition? (Select all that apply)

- i. Strategic uncertainty - Unclear threat timeline or business impact assessment incomplete
- ii. Resource constraints - Insufficient budget, personnel, or technical expertise allocated
- iii. Technology readiness - Waiting for standardisation or maturity of quantum-resistant solutions
- iv. Organisational readiness - Governance, risk management, or change management processes not yet established
- v. External dependencies - Awaiting vendor solutions, regulatory guidance, or industry coordination
- vi. Competing priorities - Other cybersecurity or digital transformation initiatives taking precedence
- vii. Risk tolerance - Current threat level assessed as acceptable for business operations
- viii. No limitations identified

Question 24. What upgrade plans exist to address IT infrastructure limitations hindering quantum security transitions?

- i. Hardware upgrades to support increased processing requirements
- ii. Network infrastructure enhancements for larger cryptographic overhead



- iii. Legacy system modernisation or replacement
- iv. Implementation of crypto-agility frameworks libraries
- v. Development of hybrid cryptographic approaches
- vi. Other plans: _____
- vii. No plans yet

Developing strategies and building capabilities to address cybersecurity risks associated with quantum

3.6 Strategy Development and Capability Building

Question 25. Have you identified the technical competencies required for your FI's quantum security transition?

- i. Detailed competency framework developed
- ii. Basic skill requirements identified
- iii. Assessment in progress
- iv. Not yet identified

Question 26. What training initiatives have you implemented for your FI's quantum security preparedness? (Select all that apply)

- i. Executive awareness programmes
- ii. Technical training for IT staff
- iii. Risk management training
- iv. Vendor management training
- v. Cross-functional workshops
- vi. External certification programmes
- vii. No training implemented yet
- viii. Others: _____



Question 27. Do you have a timeline or roadmap for transitioning to quantum-resistant cryptographic solutions?

- i. In less than 6 months
- ii. In 6 to 12 months
- iii. 1-2 years
- iv. 2-5 years
- v. Beyond 5 years
- vi. No timeline or roadmap established

Question 28. If you have a timeline, please indicate the systems that will be migrated to quantum-resistant cryptographic solutions, for both customer facing and internal systems. (Select all that apply)

- i. Customer-facing systems: ____
- ii. Internal systems: ____
- iii. Not decided which systems needs to be migrated

Question 29. If you selected “no timeline or roadmap”, please indicate the primary reasons for not having an established timeline (select all that apply):

- i. Strategic uncertainty - Unclear threat timeline or business impact assessment incomplete
- ii. Resource constraints - Insufficient budget, personnel, or technical expertise allocated
- iii. Technology readiness - Waiting for standardisation or maturity of quantum-resistant solutions
- iv. Organisational readiness - Governance, risk management, or change management processes not yet established
- v. External dependencies - Awaiting vendor solutions, regulatory guidance, or industry coordination
- vi. Competing priorities - Other cybersecurity or digital transformation initiatives taking precedence
- vii. Risk tolerance - Current threat level assessed as acceptable for business operations

Question 30. Have you allocated dedicated budget and resources for quantum security transition?

- i. Dedicated budget approved and allocated
- ii. Budget under consideration
- iii. Included in general cybersecurity budget
- iv. No specific budget allocation



Question 31. What support would be most valuable from government or industry bodies? (Select all that apply)

- i. Technical guidance and standards
- ii. Industry best practice sharing
- iii. Vendor solution certification
- iv. Training and capability building programmes
- v. Regulatory clarity and timelines
- vi. Industry-wide coordination initiatives
- vii. Funding or incentive programmes
- viii. Research and development collaboration opportunities
- ix. Threat intelligence sharing platforms

3.7 Policy and Procedure Updates

Question 32. Have you reviewed and updated internal policies to address quantum security transition?

- i. Comprehensive policy review and updates completed
- ii. Partial policy updates implemented
- iii. Review in progress
- iv. Not yet started

Question 33. Which policy areas will be updated and/or developed? (Select all that apply)

- i. Information security policies
- ii. Risk management frameworks
- iii. Third-party risk / Vendor management procedures
- iv. Incident response plans
- v. Business continuity plans
- vi. Data classification standards
- vii. Cryptographic standards



viii. Audit and compliance frameworks

3.8 Proof-of-Concept and Experimentation

Question 34. Have you conducted or planned proof-of-concept trials with quantum security solutions?

- i. Multiple trials completed or ongoing
- ii. Single trial completed or ongoing
- iii. Trials planned within next 12 months
- iv. No trials planned

Question 35. If trials have been conducted, which solutions were tested? (Select all that apply)

- i. Post-quantum cryptography (PQC)
- ii. Quantum Key Distribution (QKD)
- iii. Others: ____

Question 36. Where were these trials conducted?

- i. Singapore
- ii. Overseas

Question 37. If trials have been conducted, please share some learning points, positive outcomes or challenges.

Example:

- i. *Technical performance: e.g. slower processing speeds compared to classical cryptography, Increased computational resource requirements, larger key sizes impacting storage and transmission, network bandwidth limitations with quantum solutions*
- ii. *Integration challenges: e.g. difficulty integrating with existing legacy systems, compatibility issues with current security infrastructure, complex key management in hybrid environments, interoperability problems between different PQC algorithms*
- iii. *Implementation issues: e.g. limited vendor solution maturity, insufficient technical documentation or support, staff training and skill development requirements, extended implementation timelines*



- iv. *Operational impact: e.g. system availability concerns during transition periods, need for backup and recovery procedure modifications, monitoring and alerting system adjustments required, user experience degradation*
- v. *Cost considerations: e.g. higher than expected implementation costs, need for additional hardware upgrades, ongoing maintenance cost increases, resource allocation challenges*
- vi. *Positive outcomes: e.g. successful validation of quantum-resistant solutions, enhanced team understanding of quantum security, improved vendor relationships and support, better preparedness for future migration, valuable insights for FI-wide rollout planning*

Feedback: _____

3.9 Risk Mitigation and Contingency Planning

Question 38. Have you developed risk mitigation strategies for assets that cannot be migrated to post-quantum cryptography?

- i. Comprehensive mitigation plans developed and tested
- ii. Basic mitigation plans in place
- iii. Strategies under development
- iv. Not yet developed

Question 39. Do you have contingency plans for scenarios where quantum threats materialise earlier than predicted?

- i. Detailed contingency plans developed and tested
- ii. Basic contingency plans in place
- iii. Plans under development
- iv. No contingency plans